

Master Thesis — Progress Report

Rewiring the market clock in Spanish electricity markets

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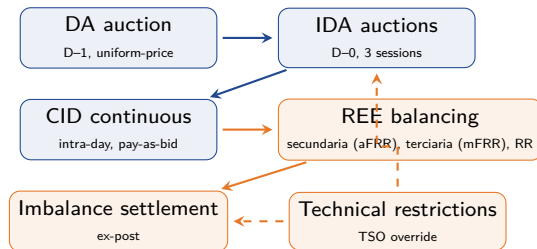
Outline

- Motivation
- Market structure and the 2024–2025 reform sequence
- Economic mechanisms: theoretical anchors and our mechanism
- What we want to test, data, and bid-shape evidence
- Identification design and main result
- What is robust, what is not, and what is left

Motivation

- Empirically assess the impact of Spain's 2024–2025 **market time unit (MTU15) reform sequence** on *firm strategic behaviour* and *market outcomes*.
- Why interesting: the reform changes the time-unit granularity of the wholesale market in three stages, with *transitional asymmetric-clock windows* where settlement and trading run on different clocks. A useful setting to study how granularity shapes firm conduct.
- Why it is hard: the reform sequence overlaps with confounders — gas-price normalisation (Dec 2023), the 2025-04-28 blackout and post-blackout *operación reforzada* (reinforced operation), the elimination of CNMC Rule 28.8 for bilateral contracts (Mar 2025).
Institutional awareness is essential.
- This presentation: identification strategy, headline result on **dominant firms** (Iberdrola, Endesa, Naturgy, EDP) strategic intraday repositioning, and an honest account of what survives robustness and what does not.

The Spanish wholesale market



- **OMIE (market operator)**: day-ahead, intraday auctions, continuous intraday.
- **REE (TSO)**: balancing markets (secundaria = aFRR, terciaria = mFRR, replacement reserve), imbalance settlement, technical restrictions.
- **Sequence**: forward markets get progressively closer to delivery; final adjustments via balancing.
- TSO can modify OMIE programmes when network or stability constraints bind.
- Firms have multiple instruments to take strategic positions across stages.

The 2024–2025 reform sequence

- Traditionally MTU = **60 min** everywhere.
- **Jun 2024 — IDA reform:** 6 local Iberian sessions consolidated into 3 European IDA sessions (still 60-min granularity).
- Three MTU15 reform stages:
 - **Dec 2024 — ISP15:** imbalance settlement → 15 min.
 - **Mar 2025 — ID15:** intraday → 15 min.
 - **Oct 2025 — DA15:** day-ahead → 15 min.
- Two transitional *asymmetric-clock* windows: settlement runs at finer granularity than trading.
- Plus: **2025-04-28 blackout** → *operación reforzada*, continuous from Apr 28 onward.

Period	ISP	ID	DA
Pre-Dec 2024	60	60	60
Dec24–Mar25	15	60	60
Mar25–Sep25	15	15	60
Oct25 onward	15	15	15

ISP15

Dec 2024

ID15

Mar 2025

DA15

Oct 2025

Theoretical anchors and our mechanism

Allaz & Vila (1993, JET) — forward markets dispel rents

Adding more forward markets prior to the spot stage commits Cournot firms to quantities and erodes withholding incentives. *Caveat:* A–V adds **more markets for the same good** (e.g., a second forward stage). Our reform **divides one good into four sub-goods** (an hour into 4 quarters). **Related intuition; not the same mechanism.**

Ito & Reguant (2016, AER) — sequential markets and arbitrage

Firms with market power reposition across sequential markets (under-commit in DA, sell more in IDA). *Caveat:* I–R is about **sequencing across markets**; we are about **granularity within a market**. **Related, not equivalent.**

Chang (2026, WP) — settlement vs dispatch period mismatch

When settlement period differs from dispatch period, averaging effects change bidding incentives. *Does not apply to DA15/ID15:* settlement and dispatch are both 15-min — no averaging. Only the ISP15-with-DA60 transition window had averaging effects, and only on *quantities* (imbalance settlement averaged within the hour).

Our preliminary mechanism — within-market granularity (what we test)

Splitting an hour into 4 quarters that clear independently lets firms bid different prices and quantities across quarters of the same hour. **Especially valuable when residual demand varies steeply within the hour** (morning ramps, evening peaks). In flat hours where within-hour demand is constant, granularity has no economic content. **To be formalized!**

What we want to test, and how

Our mechanism predicts. Dominant firms exploit within-hour granularity in *critical* hours (steep within-hour ramps in residual demand) but not in *flat* hours (constant within-hour profile). The reform should differentially affect critical vs flat hours.

Outcomes:

- q_2 = dominant firms' *voluntary* IDA repositioning per firm-day-hour (signed PIBCIE), MWh. **Headline outcome.**
- q_1 = dominant firms' day-ahead auction-cleared sell, MWh per firm-hour.
- Bid-shape: tranche count, ladder slope, qty-weighted price (post-MTU15-DA only).

How we define critical and flat hours. For each hour-of-day h , compute $\sigma_{\text{within}}^2(h)$ = **the variance of conventional production (load – wind – solar) across the four 15-min quarters of hour h** , averaged over days. High σ_{within}^2 means dispatchable units need to ramp a lot *within* the hour — exactly where within-hour granularity has economic content.

- **Treated** = critical hours $h \in \{7, 8, 16, 17, 18\}$. Top-5 by σ_{within}^2 (morning ramp + evening peak).
- **Control** = flat hours $h \in \{3, 4, 5\}$. Bottom of σ_{within}^2 (pre-dawn).

Why flat hours are a valid counterfactual. If the granularity reform has *zero economic content* in flat hours (because there is nothing within the hour to slice strategically), then comparing critical-hour behaviour to flat-hour behaviour identifies the mechanism. *This is a hypothesis I need to defend with a small theoretical model showing that, in equilibrium, the reform has no effect for flat hours — ongoing.*

Data

Coverage: 8 years (2018-01 → 2026-04), at native 60-min/15-min granularity throughout.

Three sources:

- **OMIE (Iberian market operator):** prices and *bid-level* data for DA, IDA, continuous; per-firm and per-unit programmes. ~5.7 billion rows (~413 GB raw, ~198 GB processed).
- **ENTSO-E Transparency Platform:** imbalance volumes and prices, balancing settlement, generation by type/unit, load and renewable forecasts. All system-level fundamentals come from here.
- **ESIOS (REE):** Spanish per-segment settlement, real-time prices, balancing offers. **API token requested 3 weeks ago; still pending due to technical issues on the operator's side.** Reserves attribution at firm level partially blocked.

Engineering principles:

- Pipelines are **idempotent**: re-running any step produces the same output and includes new data — supports replication.
- 25+ data families, each with its own parser and tests.
- Empirical findings tracked formally (alive / wounded / dead) with a documented audit trail.

Honest caveat. Most of the empirical findings in the audit log are descriptive rather than carrying clear economic interest — e.g., the structural break in Q1 2024 nuclear bilateral share. The presentation focuses on the small subset that has both economic content and clean identification.

Bid-shape: dominant firms exploit granularity (justifies the design)

Direct evidence at the bid level that operators *actually use* within-hour granularity post-MTU15-DA — this justifies using flat hours as a valid counterfactual.

What we observe. For dominant-firm CCGT in critical hours (post-MTU15-DA):

- More price tranches per quarter (+8% vs flat hours).
- Steeper ladder slope (+13%).
- Lower qty-weighted average price.

Interpretation. Operators bid a richer ladder in critical hours: low-price inframarginal tranches at the bottom (push the qty-weighted average down, lock clearing) plus a steeper high-price tail (capture scarcity rent). They are unambiguously exploiting within-hour granularity.

Tech selectivity. Concentrated in CCGT and (mildly) hydro. Wind, solar, nuclear show no engagement. Crucially: many plants **mechanically repeat their previous hourly bid identically across all 4 quarters** — not exploiting the granularity at all.

Why this matters for the design. Bids confirm operators *can* and *do* use within-hour granularity in the right hours. Where granularity is exploited (critical), we should see effects on quantities; where not (flat), we should not.

The DiD specification

Panel. The unit of observation is (firm f , day d , hour h). We aggregate across the 4 quarters within an hour. So the outcome is q_{fdh} .

Specification (very preliminary):

$$q_{fdh} = \alpha + \beta_1 \text{crit}_h + \beta_2 \text{post}_d + \beta_3 (\text{crit}_h \times \text{post}_d) + \gamma_f + \delta_{\text{DOW}(d)} + \varepsilon_{fdh}$$

β_3 is the DiD coefficient. γ_f are firm fixed effects (Iberdrola, Endesa, Naturgy, EDP); δ_{DOW} are day-of-week fixed effects. Cluster-robust SE by date d .

Window in each regression:

- **Pre** = pre-DA15 reform baseline: 2018-01-01 $\rightarrow$ 2024-06-13.
- **Post** = DA15/ID15: 2025-10-01 $\rightarrow$ present.
- Intermediate regimes (3-sess, ISP15-win, DA60/ID15) handled separately when needed.

Why the DiD removes time-varying confounders:

- Critical and flat hours come from the *same dates* $\Rightarrow$ daily shocks (weather, demand level, fuel prices, regulatory announcements, macro conditions) absorb mechanically in the within-day comparison.
- Pre-DA15-reform crit-flat differential serves as the parallel-trends sanity check.
- The contrast we identify is *mechanism-targeted*: hours where granularity should bite vs hours where it should not.

Heterogeneity. Most of the response carried by Naturgy (CCGT + Hydro). Iberdrola contributes through nuclear (caveat: allocation artefact). Endesa and EDP near-zero. Different technologies respond very differently — consistent with the bid-shape evidence on the previous slide.

Headline result: dominant firms reposition more in critical hours

Same panel and spec as previous slide: outcome q_{fdh} (signed IDA repositioning, MWh per firm-day-hour); $\text{crit}_h = 1$ for $h \in \{7, 8, 16, 17, 18\}$, 0 for $h \in \{3, 4, 5\}$; $\text{post}_d = 1$ from 2025-10, 0 for the pre-DA15-reform baseline.

	Estimate	SE	p-value
β_1 (crit_h , pre-DA15-reform crit–flat gap)	+41.21	3.35	1.1×10^{-34}
β_2 (post_d , flat-hour level shift pre→post)	−78.53	10.40	4.3×10^{-14}
β_3 ($\text{crit}_h \times \text{post}_d$, DiD)	+ 58.60	12.18	1.5×10^{-6}

$N = 78,232$ firm-day-hours; cluster-robust SE by date.

Type I / type II tradeoff. With $N \approx 80k$, statistical power is very high (low type II error), so the conventional $p < 0.05$ threshold becomes too lenient and inflates type I error. Worth thinking carefully about how to set significance thresholds in this regime.

Why these hours. The top-5 critical and bottom-3 flat are the hours where the robustness battery (per-hour DiD, cumulative critical-set sensitivity) confirms the effect most reliably — we use them as the cleanest contrast.

Interpretation. The outcome q_2 (signed IDA repositioning above DA commitment) is the empirical signature **Ito & Reguant (2016)** use to detect strategic intra-day arbitrage: positive q_2 means firms under-commit in DA and resell in IDA. We adopt their measure but ask a *different* question: **does this strategic IDA repositioning grow more in critical hours after the reform splits each hour into 4 sub-periods?** The DiD coefficient $\beta_3 = +59$ says yes. Mechanism: **the reform lets firms bid different prices/quantities for each quarter of an hour. This pays off where the within-hour production profile is steep — so firms exploit the ramps. Flat hours have no within-hour variation, so no effect.**

What is robust, what is not

Robust:

- The within-DA15/ID15 fact: +100 MWh more IDA repositioning in critical vs flat hours per firm-hour.
- **Fringe placebo:** same DiD on top non-dominant firms (EV, DET, EGL, EHN, REP) gives $\beta_3 = -24.27$ (SE 8.07, $p = 2.7 \times 10^{-3}$) — opposite-signed, consistent with the strategic-mechanism interpretation (mechanism requires market power).
- Bid-shape rich-ladder pattern in CCGT post-MTU15-DA.
- Two-cluster mechanism: ramp + peak positive; midday negative.

Not robust:

- The reform-attributable component (DiD δ) is a range, not a point estimate. Pre-trend instability is real.
- Conditional parallel trends with renewable + demand controls does not hold.
- Heterogeneity: Naturgy carries most of the result; cross-firm response is uneven.

The theoretical-model gap. The design needs a stylised IO model showing that, in equilibrium, the within-market granularity reform is a no-op for flat hours — this is what justifies using them as control. The model is anchored in *our* mechanism (bidding differently across the 4 quarters of an hour, exploiting within-hour ramps), not directly in Allaz–Vila or Ito–Reguant. **Out of scope:** a fully estimated structural auction model.

What is left

- **Stylised theoretical model** that justifies flat hours as control — the foundation of the empirical design.
- **Better identification strategy.** The clean flat-hour design came together yesterday; there is more thinking to do.
- Think about the demand side more — the current mechanism and design are supply-side focused.
- Include the continuous market in the analysis.
- Additional robustness checks once the mechanism interpretation is locked.
- Per-segment settlement once the ESIOS API token clears, but not necessary a bottleneck for the main result.

References

- Allaz, B., & Vila, J.-L. (1993). Cournot competition, forward markets and efficiency. *Journal of Economic Theory*, 59(1), 1–16.
- Chang, C. (2026). Settlement-period granularity and supply-function equilibrium in electricity markets. Working paper.
- Ito, K., & Reguant, M. (2016). Sequential markets, market power, and arbitrage. *American Economic Review*, 106(7), 1921–1957.

Thank you!

Appendix — robustness checks (in words)

- **Same-calendar-month restriction.** Restricting the pre-DA15-reform baseline to the same months as the post window (Oct–May) leaves the result at 96% of the magnitude. Calendar mix not the driver.
- **Baseline-window sensitivity.** Across 5 narrower pre-DA15-reform windows, the within-DA15/ID15 fact (+100) is constant; reform-attribution component ranges +27 to +133. **To think about:** pre-trend in the crit-flat gap is not stable.
- **Per-hour decomposition.** Treating each hour individually shows two clusters with opposite signs (ramps + peak positive; midday negative). Justifies the 5-hour critical set.
- **Conditional parallel trends.** Wind + solar + demand $\times$ hour controls absorb only $\sim 13\%$ of the pre-trend. *Does NOT hold.* **To think about:** an unobserved time trend (CCGT retirement, regulatory anticipation) is at play.
- **Bid-shape and extensive-margin.** Unit-level confirmation: richer bid ladders in critical hours (CCGT only); 13 CCGT units inactive in the asymmetric-clock window, return under *operación reforzada*.

Appendix — fringe placebo

Same DiD spec, run on fringe firms (top non-dominant firms in PIBCIE: EV, DET, EGL, EHN, REP). If our mechanism (within-market granularity exploited by firms with pivotal market power) is correct, fringe firms — by definition without pivotal market power — should NOT show the same response.

	Estimate	SE	p-value
Dominant firms (IB, GE, GN, EDP)	+58.60	12.18	1.5×10^{-6}
Fringe (EV, DET, EGL, EHN, REP)	-24.27	8.07	2.7×10^{-3}

β_3 ($\text{crit}_h \times \text{post}_d$) only; same outcome (q_2 , MWh/firm-hour) and same window. Dominant N = 78,232; fringe N = 46,327.

Reading. Dominant firms reposition +59 MWh more in critical hours after the reform. Fringe firms go the *opposite direction* (-24 MWh, $p < 0.01$). The mechanism asks for market power, and the placebo confirms it: where the strategic optionality of within-hour granularity is not exercisable (no pivotal market power), the response is null or opposite-signed — consistent with fringe firms being crowded out of strategic positions by the dominant firms' more aggressive critical-hour bidding.